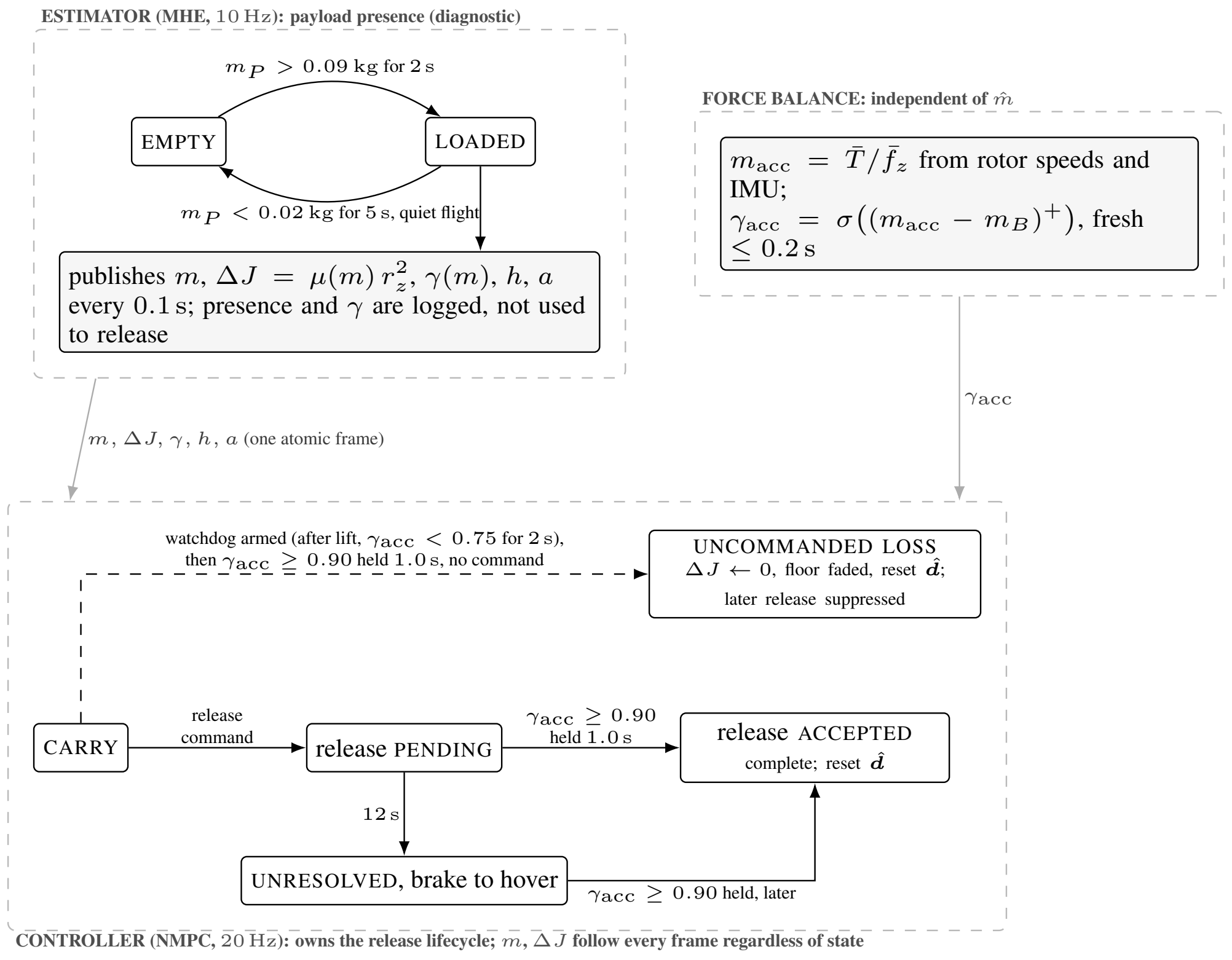

ESTIMATOR (MHE, 10 Hz): payload presence (diagnostic)
$m_P > 0.09$ kg for 2 s
EMPTY
LOADED
$m_P < 0.02$ kg for 5 s, quiet flight
publishes m, $\Delta J = \mu(m)\, r_z^2$, $\gamma(m)$, h, a every 0.1 s; presence and γ are logged, not used to release
FORCE BALANCE: independent of $\hat{m}$
$m_{\mathrm{acc}} = \bar{T}/\bar{f}_z$ from rotor speeds and IMU;
$\gamma_{\mathrm{acc}} = \sigma\big((m_{\mathrm{acc}} - m_B)^+\big)$, fresh ≤ 0.2 s
γ_{acc}
m, ΔJ, γ, h, a (one atomic frame)
watchdog armed (after lift, $\gamma_{\mathrm{acc}} < 0.75$ for 2 s), then $\gamma_{\mathrm{acc}} \geq 0.90$ held 1.0 s, no command
UNCOMMANDED LOSS
$\Delta J \leftarrow 0$, floor faded, reset $\hat{\boldsymbol{d}}$;
later release suppressed
CARRY
release command
release PENDING
$\gamma_{\mathrm{acc}} \geq 0.90$ held 1.0 s
release ACCEPTED
complete; reset $\hat{\boldsymbol{d}}$
12 s
UNRESOLVED, brake to hover
$\gamma_{\mathrm{acc}} \geq 0.90$ held, later
CONTROLLER (NMPC, 20 Hz): owns the release lifecycle; m, ΔJ follow every frame regardless of state